

A Systematic Review of Machine Learning Models Against MIMIC and Currency Demand Models on Estimating the Size of Informal Economy

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Abstract—Estimating the size of the informal economy remains a persistent methodological challenge due to its hidden and unregulated nature. Traditional econometric approaches, particularly the Multiple Indicators Multiple Causes (MIMIC) model and the Currency Demand Approach (CDA), have long dominated the literature, yet both rely on restrictive assumptions that may fail to capture the nonlinear and heterogeneous characteristics of informal economic activity. Recent advances in artificial intelligence (AI) and machine learning (ML) offer new possibilities for improving estimation accuracy and flexibility. This systematic review critically examines whether and how machine learning models enhance or outperform traditional econometric methods, specifically MIMIC and CDA, in estimating the size of the informal economy. Following the systematic literature review protocol established by Kitchenham (2004), a structured search was conducted across Scopus and Web of Science databases for publications between 2010 and 2025. After screening and quality assessment, 15 studies were selected for final synthesis. Data were extracted and analyzed using a qualitative thematic framework. The findings reveal four primary mechanisms through which ML models enhance or outperform traditional methods. Machine learning models outperform traditional econometric methods in predictive accuracy, non-linearity handling, spatial learning, and mixed-data integration. However, MIMIC and CDA remain superior for causal interpretation and hypothesis testing.

Keywords—Informal economy, shadow economy, MIMIC model, currency demand approach, machine learning, artificial intelligence

I. INTRODUCTION

The informal economy refers to economic activities that fall outside the reach of government regulation, taxation, and formal oversight. No country is exempt; its presence ranges from street vendors and small-scale traders to undeclared cash transactions between employers and workers. Several structural conditions feed into it: bureaucratic barriers, heavy tax burdens, poverty, unemployment, weak governance, corruption, and exclusion from formal financial systems [4]. In many developing countries, the formal economy simply cannot generate jobs fast enough to absorb the available labour force.

Measuring the informal economy typically means expressing it as a share of Gross Domestic Product (GDP) or total employment. This captures unregistered enterprises, irregular work arrangements, and income that goes unreported to authorities. Its size is shaped by a combination of variables like tax pressure, institutional capacity, corruption, joblessness, and household income levels. Where

taxes run high and institutions are fragile, the informal sector tends to be larger. Estimation methods fall into two broad categories: direct approaches, which rely on household or firm-level surveys, and indirect ones, which infer activity from currency circulation patterns or composite indicator models. Machine learning has entered this space as well, improving predictive fit, though estimates remain sharply sensitive to whichever methodology a researcher chooses [6].

This review asks a focused question: Do machine learning models actually improve on or outperform established econometric tools when it comes to estimating informal economy size? The two benchmarks under scrutiny are the MIMIC model and the CDA. Drawing on a systematic reading of existing research, the analysis maps out how these methods differ in structure, what variables each one prioritizes, and how country context shapes results. Beyond that comparison, the review works to pin down the strengths and blind spots of each approach, examine whether estimates produced under different frameworks can meaningfully be compared, and flag the empirical gaps that future research needs to address

II. BACKGROUND

Measuring the informal economy remains a persistent methodological challenge in economic research due to the hidden and unregulated nature of informal economic activity. Traditional econometric approaches, particularly the MIMIC model and CDA, have long dominated the literature; however, both methods rely on restrictive assumptions that may fail to capture the heterogeneous and nonlinear characteristics of informal economic activity across diverse country contexts [7]. With the growing availability of artificial intelligence (AI) and machine learning (ML) techniques, researchers have increasingly explored whether data-driven approaches such as Random Forests and Support Vector Regression (SVR) can provide more accurate and flexible estimation frameworks [5]. Although previous reviews have examined either traditional econometric methods or emerging ML applications in isolation, the existing evidence remains fragmented across methodologies, datasets, and regions, with no comprehensive synthesis that systematically compares ML approaches with traditional econometric models and evaluates their respective strengths, limitations, and applicability in the specific context of informal economy estimation. This gap motivates the present review.

III. RESEARCH METHODOLOGY

A. Review Protocol

The review is grounded in Kitchenham's [16] three-phase protocol — planning, conducting, and reporting.

Planning meant nailing down the research aims and deciding which studies would make the cut and which would not. The conducting phase involved a literature search, record screening, quality checks, and data extraction from studies that met the criteria. Those findings then fed into a qualitative synthesis that shaped the review's conclusions.

Staying within this structure keeps the work transparent, repeatable, and less vulnerable to bias in how studies were chosen.

B. Research Question

In accordance with Kitchenham's [16] SLR framework, this review was structured to investigate the following research question:

RQ: How do machine learning models enhance or outperform traditional econometric methods, particularly MIMIC and CDA, in estimating the size of the informal economy?

C. Data Sources and Search Strategy

To identify relevant literature, two major peer-reviewed academic databases were used: Scopus and Web of Science (WoS). These were selected for their broad multidisciplinary coverage, rigorous indexing, and reliability in supporting systematic literature reviews. The search was limited to publications from 2010 to 2025, reflecting the period during which ML methods began gaining meaningful traction in economic and social science research.

The search strategy was built around four primary keywords.

1) *“informal economy size” AND “machine learning” for SCOPUS*

2) *“informal economy size” AND “MIMIC” for WoS*

3) *“informal economy size ” AND “currency demand approach” for WoS*

4) *“informal economy size” AND “machine learning” for WoS*

The results from all four search strings were combined across both databases, and duplicate records were removed. The remaining records were screened by title and abstract for relevance to the research question. Following this initial screening, 42 papers were retained for full-text review and evaluated against criteria of methodological rigour, empirical validation, and direct relevance to the topic. After full-text assessment, 15 studies satisfied all inclusion criteria and were selected for the final synthesis, representing the most methodologically robust and thematically relevant contributions to the field.

D. Inclusion and Exclusion Criteria

To obtain a manageable and reliable set of studies, a structured screening process was applied based on Kitchenham's systematic review guidelines [16]. The research question at the heart of this review served as the main reference point whenever a decision was made about whether a particular study should be included or excluded.

Well before any papers were looked at individually, a working set of criteria was put together to make those calls as consistent as possible. For a study to make the cut, it had to be both empirical and peer-reviewed, and it had to deal directly with at least one of three things: applying ML to the measurement of informal or shadow economic activity, comparing machine learning tools against econometric approaches like the MIMIC model or the CDA or pushing forward how unobserved economic activity gets measured through data-driven means. On top of that, the work itself had to be methodologically transparent, backed by real empirical findings, and genuinely relevant to what this review was trying to answer.

Anything that did not tie back clearly to estimating the scale of the informal economy was left out. That meant setting aside studies where ML was being used for economic problems with no connection to informality, where MIMIC or currency demand methods were applied in completely unrelated settings, or where the informal economy was only ever discussed in qualitative or policy terms with nothing empirical to show for it. Work that treated machine learning or econometric modelling purely on its own terms, without any real grounding in informal economy research, did not fit the scope of this review and was excluded accordingly.

1) Primary Selection:

The opening stage of selection was intentionally light, resting on just two basic conditions before anything else came into play:

- Papers had to be written in English
- Publication had to fall **within 2010 and 2025** — This period was chosen because it corresponds to the rise of machine learning and deep learning in economics and social sciences, as well as growing recognition of the limitations of MIMIC and CDA.

Studies that did not meet either criterion were removed at this stage to ensure that the later screening process was not slowed down by irrelevant material.

2) Secondary Selection

Studies passing the primary screening stage were subsequently assessed based on their titles and abstracts. This stage aimed to determine whether each study was substantively relevant to the research question. A record was retained if its title or abstract indicated a clear focus on at least one of the following:

- Work that puts ML tools to use in estimating, modelling, or predicting how large the informal or shadow economy actually is.
- Any kind of direct engagement or head-to-head comparison with traditional econometric tools — the MIMIC model and CDA in particular, in the context of measuring shadow economic activity.
- Efforts to develop or test data-driven ways of getting at the informal economy size, whether as a standalone approach or as something meant to sit alongside existing indirect estimation methods.

Studies whose titles and abstracts revealed an exclusive focus on unrelated econometric applications, general machine learning benchmarking outside economic contexts, or informal economy topics addressed through purely qualitative or policy-oriented lenses were excluded at this stage.

3) Full-text Assessment

Papers that made it through the second stage were then pulled up in full and gone through carefully. Each one was held up against the complete set of criteria, whether the empirical work held up, and how directly the study actually contributed to answering the question this review was built around. Three things were looked for in particular:

- Whether the study had a clearly laid out ML methodology that was genuinely put to work in estimating informal economic activity, not just mentioned in passing.
- Whether the empirical findings were substantial enough to be meaningfully set against what traditional approaches like MIMIC or CDA have historically produced.
- Whether enough methodological detail was provided to make data extraction and synthesis a realistic and reliable exercise.

Following full-text review, 15 studies satisfied all inclusion criteria and were selected for the final synthesis, representing the most methodologically robust and thematically relevant contributions to the field.

E. Study Quality Assessment

The quality of the included studies was evaluated by examining several key aspects relevant to comparative research between machine learning and traditional econometric methods for estimating the informal economy. Several things were looked at when going through each study. To begin with, how clearly the research objectives were spelt out mattered particularly whether the study was trying to get better predictive results, get to grips with nonlinear relationships, or sharpen the structural interpretation of informal economy estimates. Closely tied to this was the question of how transparently the methods were laid out, covering everything from the choice of model through to how the data was prepared and what assumptions the whole approach was resting on.

A fair amount of attention also went to the datasets themselves. Things like how many observations were included, how far back the data stretched, which countries were covered — Romania [2], Algeria [7], Vietnam [8], Peru [9], Nigeria [10], the Balkans [11], Sweden [6] among others — and how the authors dealt with gaps or noise in the macroeconomic indicators all fed into the assessment.

How the results were measured and reported was another consideration. For predictive models, metrics like RMSE, MAE, and R-squared were looked at; for MIMIC-based work, fit indices such as RMSEA, CFI, and SRMR were the relevant benchmarks; and for currency demand studies, residual diagnostics carried most of the weight. Finally, whether the work could realistically be reproduced was factored in. This includes checking for details on parameter tuning and cross-validation in the ML studies [1], [2], [5], and for sensitivity checks in the traditional ones, such as varying

the currency ratio baseline [15] or the zero-informality base year assumption [14].

The depth and coherence of the result interpretations were evaluated, particularly how researchers explained, compared, or resolved differences in results produced by different models or methods, for example, where MIMIC suggested a declining informal sector while ML predicted an increase [11].

The extent to which authors acknowledged the limitations of their work was also assessed. This included recognition of MIMIC's identification problems and sensitivity to model specification [7], [10], currency demand assumptions about constant velocity of money [15] and the absence of non-monetary transactions [14], as well as machine learning issues such as overfitting in small samples [8], lack of causal interpretability [5], and data leakage between training and test sets [4].

Finally, the practical value of the findings for real-world applications was considered, such as policy design for tax administration, anti-corruption strategies [4], labour market regulation [3], and cross-border informal activity monitoring [6]. Together, these criteria enabled a consistent and meaningful assessment of the methodological strength of the studies included in this review, distinguishing those that merely applied techniques from those that provided robust, comparative evidence on when and how machine learning enhances or outperforms traditional econometric methods.

F. Data Extraction and Synthesis

A uniform strategy was used to systematically extract information from the 15 included studies. For each study, the following was recorded: who wrote it and when, which country or region the analysis was based in and what method was actually used to arrive at informal economy estimates. Methods were sorted into two broad categories: those rooted in traditional econometrics and those drawing on ML tools. Additionally, the datasets and their key indicators, such as tax burden, regulatory quality, GDP per capita, currency ratios, unemployment rates, and shadow economy proxy variables, were systematically recorded, along with the evaluation metrics used for validation (e.g., RMSE, MAE, R-squared, goodness-of-fit indices, residual diagnostics).

Pulling everything together for synthesis meant grouping the studies thematically rather than treating each one in isolation. Three broad clusters emerged naturally from the material:

1. Traditional Structural Methods — this group brought together studies built around MIMIC models [7], [8], [9], [10], [11] and those working within currency demand frameworks [12], [13], [14], [15].
2. Machine Learning Enhancement Methods — here the focus was on studies putting random forests, gradient boosting, support vector machines, and neural networks to work on informal economy estimation [1], [2], [3], [4].
3. Advanced Deep Learning and Spatial Methods — covering deep learning architectures [5] and graph-based or neighbour-aware algorithms that incorporate spatial dependencies across neighbouring countries [6].

In addition, several application domains were considered within the informal economy literature, such as cross-country comparative analysis [1], [11], single-country time-series estimation [2], [7], [8], [9], [10], [13], [14], [15], policy-oriented economic cost assessment [4], and spatial dependence analysis [6].

After that, patterns having to do with predictive performance, validation strategies, and methodological limitations were recognized and scrutinized. Key performance patterns included the consistently higher predictive accuracy of machine learning models over traditional MIMIC and CDA, particularly when handling non-linear relationships [1], [5] and high-dimensional indicator spaces [2]. Validation patterns revealed that traditional methods rely heavily on in-sample goodness-of-fit and sensitivity analyses [7], [10], whereas ML studies employed cross-validation, train-test splits, and out-of-sample forecasting exercises [1], [3], [5]. Limitations patterns included the lack of causal interpretability in deep learning models [5], the identification problems and specification sensitivity inherent to MIMIC [7], [10], the strong assumptions underlying currency demand approaches (e.g., constant velocity of money, absence of non-monetary informal transactions) [14], [15], and the risk of overfitting when ML models are applied to small-sample country data [8].

This review examined different methods used to estimate the informal economy and explored the challenges researchers and policymakers face when choosing between them. These insights highlight the strengths and limitations of this rapidly expanding field, particularly in guiding future work toward hybrid models that combine the interpretability of econometric methods with the predictive power of machine learning.

IV. RESULTS AND DISCUSSION

A. Quality Assessment Result

Quality assessment across the 15 selected studies pointed to a solid methodological base in the informal economy estimation literature. Out of these papers, 13 scored high or excellent, with clear research objectives, well-described methods, appropriate data choices, and defined evaluation metrics. Among the stronger studies were those using deep learning architectures to measure unregistered economic activity — reporting RMSE, R^2 , and MAE to back up model accuracy [3]; machine learning techniques tested against structural equation models for the Romanian shadow economy, with cross-validation and feature importance analysis bolstering methodological credibility [2]; and spatially-informed ML frameworks drawing on high-quality Swedish register data to infer informal economy size from neighboring country characteristics [6].

On the traditional econometric side, high-quality studies included a post-transition structural estimation of Poland's informal sector with transparent specification and sensitivity analysis [14], a panel econometric analysis of regulatory intensity as a driver of informality in developing countries using fixed-effects estimation [12], a multi-country MIMIC application tracing the informal economy across the Balkans from 1996 to 2021 [11], and a cash demand econometric model for Serbia incorporating time-series diagnostics and

multiple robustness tests [15]. The remaining two studies addressing Vietnam's informal economy and Pakistan's underground economy through mixed qualitative-quantitative approaches received moderate quality ratings, primarily due to underspecified model selection criteria, limited evaluation metrics, and insufficient integration of methodological strands.

B. Research Question Results

Machine learning models enhance or outperform traditional econometric methods (MIMIC and CDA) in estimating the size of the informal economy through four primary mechanisms:

1. **Handling Non-Linear and Complex Interactions:** Unlike traditional MIMIC models, which assume linear causal structures (e.g., [7]; [10]; [9]), machine learning algorithms (e.g., Random Forest, Neural Networks) automatically capture non-linear relationships and high-dimensional interactions among causes and indicators of the shadow economy without requiring pre-specified structural equations. Research has shown that deep learning models can uncover hidden patterns in unregistered economic activity that traditional linear models may fail to detect [5].
2. **Superior Predictive Accuracy and Generalization:** Traditional currency demand approaches (e.g., [15]; [14]) and [12]) rely on historical residual comparisons and suffer from assumptions about velocity of money and baseline periods. In contrast, ML models, particularly ensemble methods and support vector machines, demonstrate lower out-of-sample prediction errors. Studies have found that ML algorithms can significantly reduce forecasting bias when estimating the size of the informal economy across multiple countries [1]. Similarly, evidence from Romania suggests that ML techniques outperform traditional time-series models by better capturing structural breaks and behavioural shifts [2].
3. **Incorporating Spatial and Neighbour Effects:** Traditional MIMIC models usually analyze each economy separately or use simple cross-country comparison variables. However, [3] and [6] demonstrate that ML methods, especially graph-based and neighbour-aware algorithms, can directly model spatial dependencies like estimating a country's informal economy based on neighbouring countries' characteristics. This outperforms the MIMIC approach's inability to incorporate such relational data seamlessly.
4. **Integration of Mixed-Data and High-Frequency Indicators:** Traditional methods rely on aggregated macroeconomic time series (e.g., GDP, tax revenue, currency ratios) with low frequency. ML models can integrate qualitative survey data [13] and high-frequency proxy variables. Research has shown that combining econometric residual estimates with ML classification techniques applied to corruption data can produce more robust estimates of economic costs than relying on either method independently

[4]. Studies also note that although the MIMIC model offers stronger structural interpretation, it is less capable of adapting to rapidly changing informal sector dynamics, whereas machine learning models can be retrained more flexibly to capture evolving patterns [8][11].

However, a key trade-off is identified: Traditional econometric methods (MIMIC, CDA) offer superior structural interpretability and hypothesis testing regarding causal determinants (e.g., tax burden, regulation, unemployment). ML models often perform better in prediction and identifying complex patterns, but their decision-making processes can be difficult to interpret, and they require careful calibration to avoid overfitting, especially when smaller datasets are used (e.g., [8], [9], [10], [11]).

V. RESEARCH GAP AND FUTURE DIRECTION

Despite evidence that ML models often outperform traditional econometric methods in estimating the informal economy, several critical gaps remain. First, the majority of research focuses on single countries or narrow regions (e.g., Romania, Algeria, Vietnam, the Balkans), with limited cross-validation across diverse economic contexts. Second, there is insufficient methodological guidance for applying ML techniques to small-sample or low-data-quality settings, where overfitting is a serious risk. Third, ML models excel at prediction but cannot provide causal estimates of how policy changes (e.g., tax increases) affect the informal economy—a critical need for policymakers. Finally, the field has underutilized temporal deep learning architectures (e.g., LSTMs, transformers) and neglected novel data sources such as satellite imagery, mobile phone records, and high-frequency transaction data.

Future research should focus on developing hybrid models that combine the structural strengths of traditional econometric methods with the predictive capabilities of ML techniques. More multi-country comparative studies and methodological innovations for small-sample settings are needed to improve the reliability and generalizability of informal economy estimates across different contexts. Researchers should also explore causal and explainable AI approaches, temporal deep learning, and alternative data sources such as nighttime lights and mobile metadata to enhance both predictive accuracy and policy relevance while promoting reproducible and interpretable estimation frameworks.

VI. CONCLUSION

This systematic review finds that ML models consistently enhance or outperform traditional econometric methods, MIMIC and CDA, in estimating the size of the informal economy through superior predictive accuracy, non-linear relationship handling, and integration of mixed-data sources. ML models demonstrate lower out-of-sample prediction errors compared to MIMIC and CDA when validated using metrics such as RMSE and MAE [1], [2], [5]. Advanced approaches further outperform traditional methods by directly modelling spatial dependencies, such as estimating a country's informal economy based on neighbouring country characteristics [3], [6]. That said, traditional econometric methods still hold the edge when it comes to causal inference and structural interpretation. They let researchers actually test hypotheses about what drives informality, whether tax pressure, regulatory quality or any other factors. ML models work differently, built around prediction, they offer little transparency into why they produce the outputs they do, which makes drawing actionable policy conclusions from them a harder task. Additionally, ML methods risk overfitting, whereas traditional methods rely on strong assumptions about currency velocity and baseline periods [8], [14], [15]. The review highlights hybrid approaches that combine the structural identification strengths of the MIMIC model with the residual prediction capabilities of ML as an emerging best practice in informal economy estimation [4]. Persistent gaps include a lack of multi-country comparative studies, insufficient guidance for small-sample ML applications, underutilization of temporal deep learning architectures like LSTMs, and neglect of novel data sources such as satellite imagery and mobile phone records. Future research should prioritize hybrid model development, causal and explainable AI, and reproducible benchmarks to advance the field beyond isolated methodological comparisons.

VII. AUTHOR CONTRIBUTION

The authors confirm the following contributions: study conception and topic analysis were carried out by Jinsy and Neethu; data collection, screening, and analysis were conducted by Jinsy and Neethu; and interpretation of results, along with manuscript writing, was completed by Jinsy and Neethu. Both authors reviewed and approved the final version of the manuscript.

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List of Abbreviations

Abbreviations	Full Form
AI	Artificial Intelligence
CDA	Currency Demand Approach
CFI	Comparative Fit Index
GDP	Gross Domestic Product
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MIMIC	Multiple Indicators Multiple Causes
ML	Machine Learning
RMSE	Root Mean Square Error
RMSEA	Root Mean Square Error of Approximation
RQ	Research Question
SLR	Systematic Literature Review
SRMR	Standardized Root Mean Square Residual
SVM	Support Vector Machine
SVR	Support Vector Regression